

E-tivity 4: The EU AI Act and developing a code of conduct for AI industry professionals



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1 An AI code of conduct for computer engineers working on ITS

1.1 Introduction

This Code is not intended to be an exhaustive statement of the rules applicable to using AI, but a call for conscious use of AI tools. It is intended for computer engineers working on *Intelligent Transport Systems* (ITS) with the authority *Transport Infrastructure Ireland* (TII) in a time when regulations are trying to catch up with technological innovations. All principles present in Fig. 1.1 will be explained. This code of conduct will be updated from time to time to reflect changes in the law, best practices, and other relevant factors; think about it as a ‘living document’.

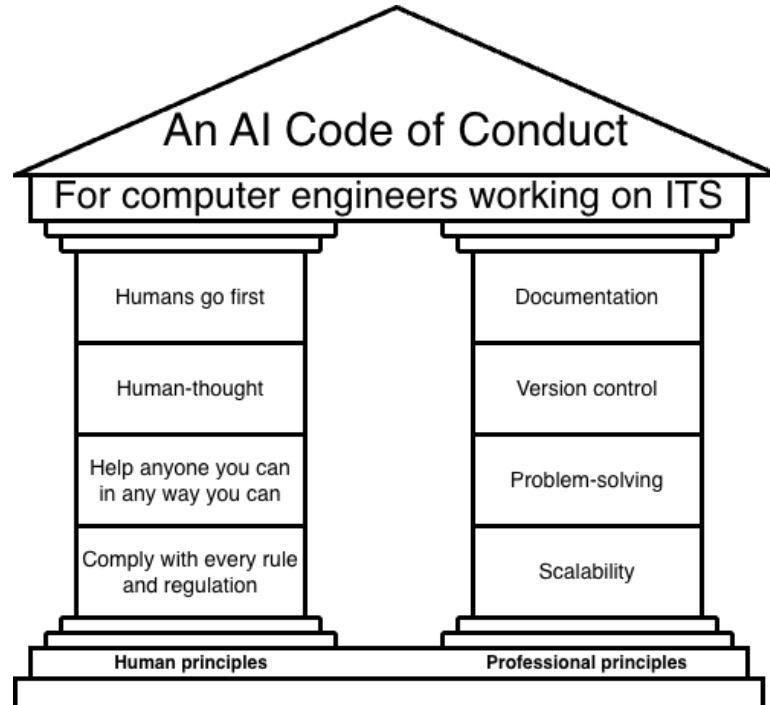


Figure 1.1: The pillars of the code of conduct - A graphic representation.

1. AN AI CODE OF CONDUCT FOR COMPUTER ENGINEERS WORKING ON ITS

1.2 Human principles

This code of conduct takes a human-centric approach that favours accountability and collaboration.

Humans go first: Traffic management and incident response will undoubtedly have to deal with human casualties or potential casualties; your work must always prioritise keeping that number as close to zero as possible. AI will return empty apologies and carry on after a mistake, but for you, the weight of a human lost will never truly leave your conscience. Always keep this as your main concern.

Human-thought: No machine can truly assess the value of a human life; the solutions you create should start and be signed off by humans. AI tools might help with content generation if the content is first fully outlined by humans and then fully reviewed by humans afterwards. AI will confidently lie to you; this flaw in the technology must not compromise the integrity of your system.

Help anyone you can in any way you can: just as you are an expert in your domain, the AI might lose valued interactions or behaviours only witnessed by you. **Do not work in a vacuum.** Hours of CCTV footage or conversations with road maintainers or first responders could reveal insights that are beyond the communication channels an AI can and should oversee. Having said that, if AI offers ways to improve how much you can help, as long as you are responsible, there is nothing to fear about its usage.

Comply with every rule and regulation: AI does not need to wear PPE or undergo health and safety training, but you do. It is also your responsibility to understand and comply with regulations such as GDPR, Datex II, EU AI Act, SAI, and any other. If more education or training is required to fully this, it is your responsibility to find the right help.

1.3 Professional principles

It is expected that you will continue to exercise good practices.

Documentation: both external (documents, diagrams...) and internal (code comments) are vital to maintain and share the understanding of your system. For any of your work, it would be your voice, clear and concise. AI coding comments are often easy to spot and deviate from expectations. AI-generated images can contain mistakes; if AI is used for visual media, the output should be editable. And **always** indicate where AI was used. Be particularly wary of involving AI hardware design, rely on suppliers and their official documentation instead.

Version control: vital to the structure of any project, committing often and with meaningful messages not only shows the efforts of your labour but also provides traceability and accountability. Being clean and organised with your coding branches and explaining the details of your implementation in pull requests is expected of you. Never leave this in the control of the AI; conversely, when possible, set up guardrails to avoid agents from controlling tools like Git.

Problem-solving: As a computer engineer, you are the one in charge of arranging, planning, and designing hardware and software solutions for traffic management authorities. AI is not a peer engineer with similar problem-solving skills to yours; it is an instrument with very specific limitations. You should use the tool when it fits (morally, lawfully and technically) but never let it atrophy your most valuable asset.

Scalability: Your solutions should not be static but allow for shrinkage or growth when needed. AI tools have learned about human natural language immensely in recent years, but this does not transfer to an understanding of the real world. You are the one in charge of the designs that an AI might help you with and it is your responsibility to ensure they are capable of handling as little or as much as the system might require at different stages of the software development process.

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1.4 Misuse, underuse, and risks

Prohibited:

- using any free tier of any AI tool, regardless of how good it looks in the demo
- Consumer tier of any AI tool, regardless of how convenient it is
- Open-weights models running locally for anything except engineering work
- AI features inside third-party tools whose terms have not been reviewed
- Any tool not approved by your organisation (if applicable)
- No work data, customer information, internal documents, or code goes through a personal AI account, ever.
- No screenshots of work artefacts, no transcripts of internal conversations. Not even half-redacted prompts.
- Never paste customer PII (personally identifying information) into any AI tool
- Never paste financial information into any AI tool
- Never paste credentials, API keys, tokens, certificates, or internal URLs
- Never paste verbatim customer documents, contracts, or settlement files
- Never use a non-approved tool for work output, even just to see what it does

This long list, although scary, should not deter you from enjoying the commodities of a virtual assistant; just consider:

- Always use the approved tier with audit logging enabled
- Always disclose AI-assisted output to the recipient when the output is customer-facing

- Always verify every named fact — person, number, citation, product — against an authoritative source before sending
- Always report a paste mistake (credentials, secrets, private information, etc) within the same working day

1.5 Raise your voice: when and how

Having understood all previous sections, the only question left is ‘What to do if you see someone breaking this code of conduct?’. This is not a light matter; misuse of AI technologies could lead to serious consequences. And in a collaborative project, even if you comply with all guidelines, one rogue individual is enough to contaminate the project. Because pressure can be external or internal, it is a priority for this code of conduct to offer relief and various levels of aid for whistleblowers. Relevant information is defined in the Protected Disclosure Act 2014 as:

- I. *in the reasonable belief of the worker, it tends to show one or more relevant wrongdoings, and*
- II. *it came to the attention of the worker in connection with the worker’s employment*

You should go through the levels in the tunnel from the top, as represented in Fig. 1.2. If the issue can be resolved with a conversation with your manager or the system’s manager, that is ideal. AI and its ability to assist to the level we experience today is only recent, and peers could not be up to speed with the good practises. However, if you suspect the issue could be rooted deeper in your project, a different figure from management might be a better point of contact. And finally, if the situation seems to be organisational, in Ireland you can make a *protected disclosure* to the Workplace Relations Commission (WRC).

1. AN AI CODE OF CONDUCT FOR COMPUTER ENGINEERS WORKING ON ITS

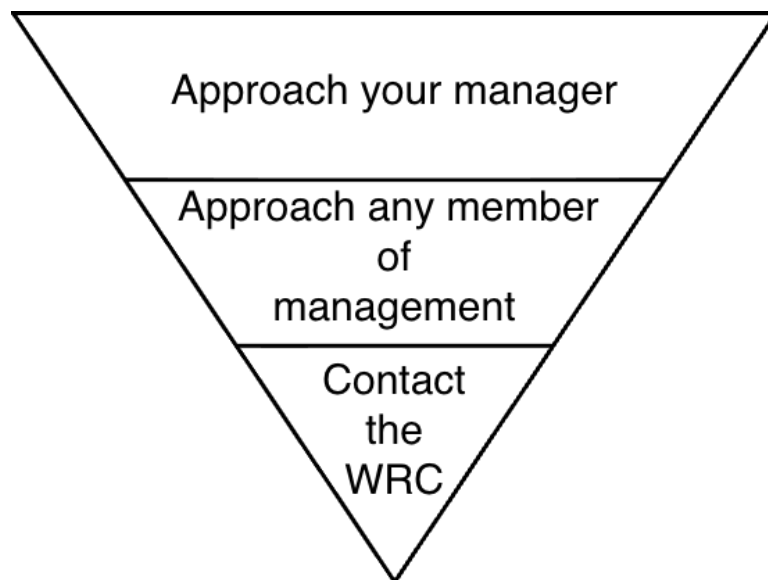


Figure 1.2: Escalation funnel - a levelled guide to seek for help.

2 Part B

This analysis of my code of conduct, by default, lives in the same space selected for the former. To defend my decisions in part one, I must explore the theory behind modern codes of conduct, how other professions have written them in the past, the recent efforts in regulating AI either with law or guidelines and also specific considerations in the area of Intelligent Traffic Systems to lastly close down with the challenges and final thoughts.

2.1 Code of conduct theory

Kvalnes (2019) introduced the concept of *moral dissonance* and left clear that the outdated view of individuals needed to change. The idea that the common individual values integrity over anything else and that wrongdoing carried on by crooked people far from the norm is dangerous because it can invalidate and block the path towards real solutions. Kvalnes explained corruption and misbehaviour as the outcome of an environment rather than a factor linked to the character's moral values.

This is why I intended my code of conduct to start with values that reminded computer engineers what is truly at stake while developing ITSs, which is human safety. The intent was to appeal to empathy, or, if this is just as naive as the integrity approach, the self-preservation instinct. To decide the four pillars of human values (section 1.2), I used Kvalnes' wheel of navigation Fig. 2.1. I focused on the aspects **law**, **ethics**, **morality** and **identity**. However, this does not mean that the other were neglected but addressed in professional principles.

Incredibly useful for structure and inspiration were the following codes of conduct:

- *Guide to Professional Conduct and Ethics for Registered Medical Practitioners* (Medical Council 2019).
- *Code of Practice for Practising Barristers* (Legal Services Regulatory Authority 2024).

2. PART B



Figure 2.1: Kvalnes' Navigation Wheel - a tool for decision makers

- *ACM Code of Ethics and Professional Conduct* (Association for Computing Machinery 2018).

2.2 Computer engineering

In order to decide which elements of the profession should be highlighted and preserved in light of the AI advancements, it was necessary to review the literature that shaped not only me but dozens of other engineers.

Fig. 2.2 encapsulates the software development process. It was the true north I wanted to always steer towards. And while seeking to find the bare-bones bricks of the professionalism pillar, I also reminded myself of many acronyms that are always useful to remember in the guild. Such as: SRP - *Single Responsibility Principle* and DRY - *don't repeat yourself* (Martin 2009). Any coder probably has at least one memory of simplifying things down following Occam's razor principle or the infamous KISS acronym.

In comparison, one inspection of 'vibe-coding' approaches like the ones shown in Fig. 2.3 and Fig. 2.4 reveals a gap that one simply cannot afford in the domain of critical infrastructure like ITSs are.

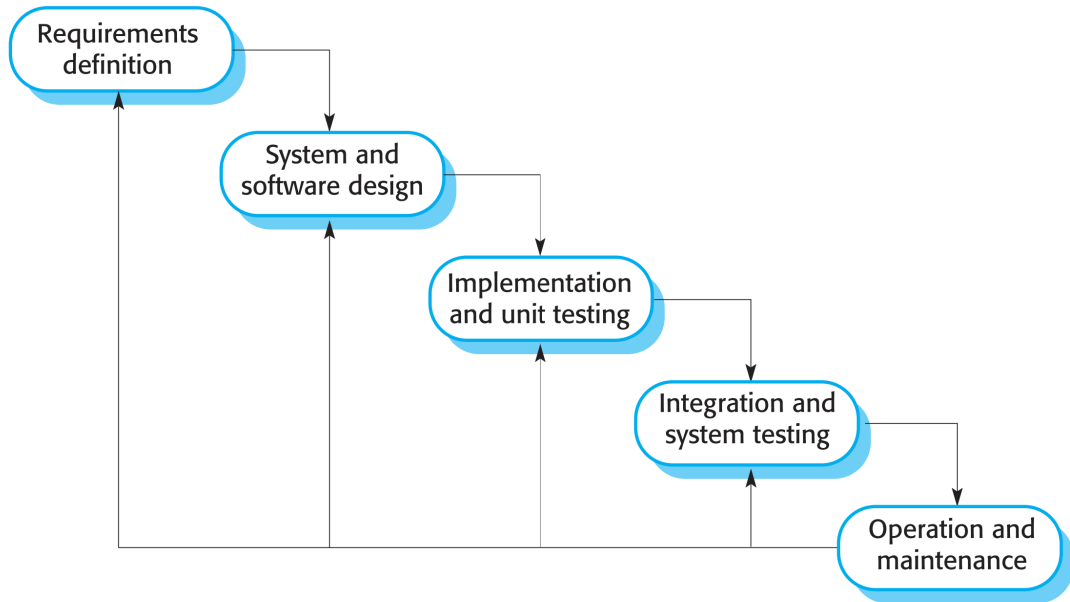


Figure 2.2: Software development process - waterfall approach as an example (Sommerville 2016).

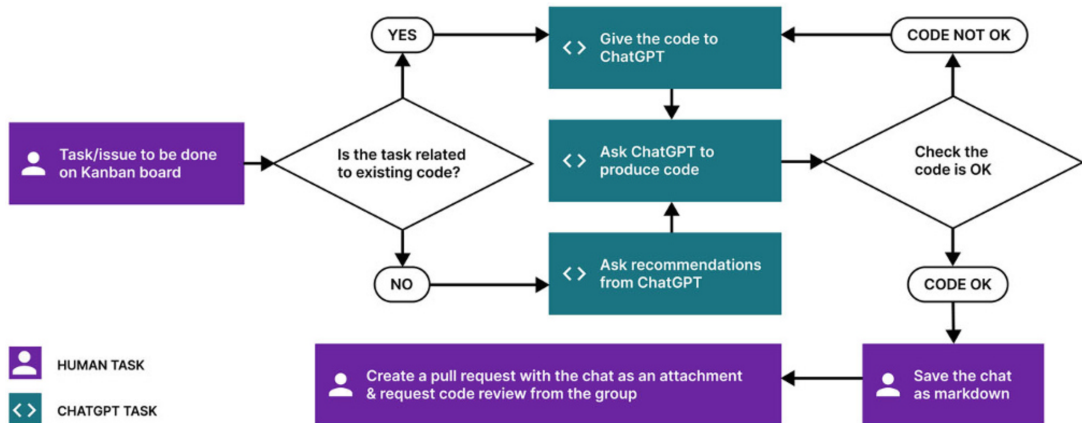


Figure 2.3: AI-Assisted Software Development - Working with ChatGPT (Liukko et al. 2024).

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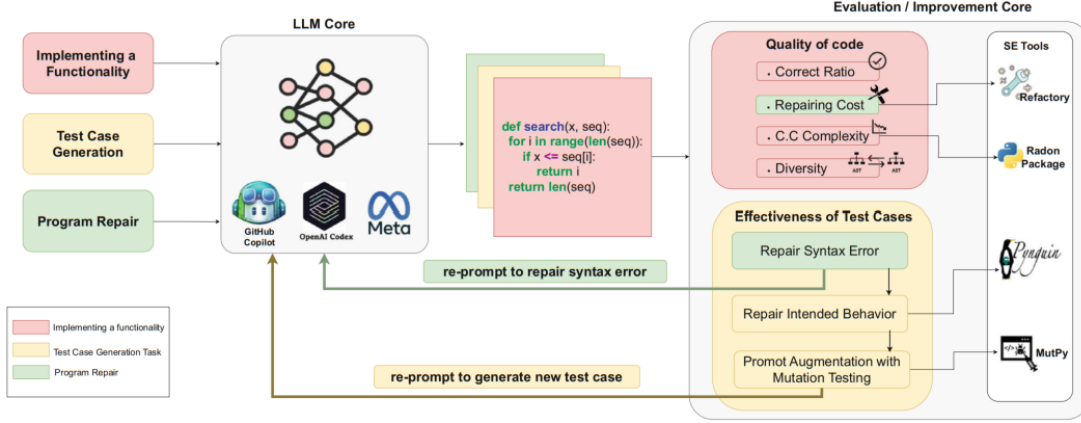


Figure 2.4: LLMs in the software development process - attempting to use AI to help in the stages of implementation, testing and maintenance (Moradi Dakhel et al. 2024).

The truth is that we are not at the highest level of automation yet 2.5, and if research has a space to carry on and keep testing the limits of each level; it is not wise to rush ourselves towards this level of automation for high-risk AI systems.

I acknowledge that the studies I referenced to exemplify what is infamously called ‘vibe-coding’ are not the most recent or the best examples of AI integration with the development process. But it is a reflection of approaches that are still present in society and need to be kept outside critical infrastructure systems.

2.3 Artificial intelligence and codes of conduct

To justify the reasoning behind my selection of principles in the context of AI, I analysed three different resources. IBM has a great piece for AI designers; the chapter on ethics was incredibly useful (International Business Machines Corporation 2022). It bases its work on IEEE’s “Ethically Aligned Design”. IBM AI Design ethical principles: (1) Accountability, (2) Value alignment, (3) Explainability, (4) Fairness and (5) User data rights

Floridi (2019b) introduced concepts in the digital ethics realm that were also of great value to develop critical thinking. And despite focusing on “risks of being unethical”, it also suggests an ethical framework. Ethics Guidelines for

2.3 Artificial intelligence and codes of conduct

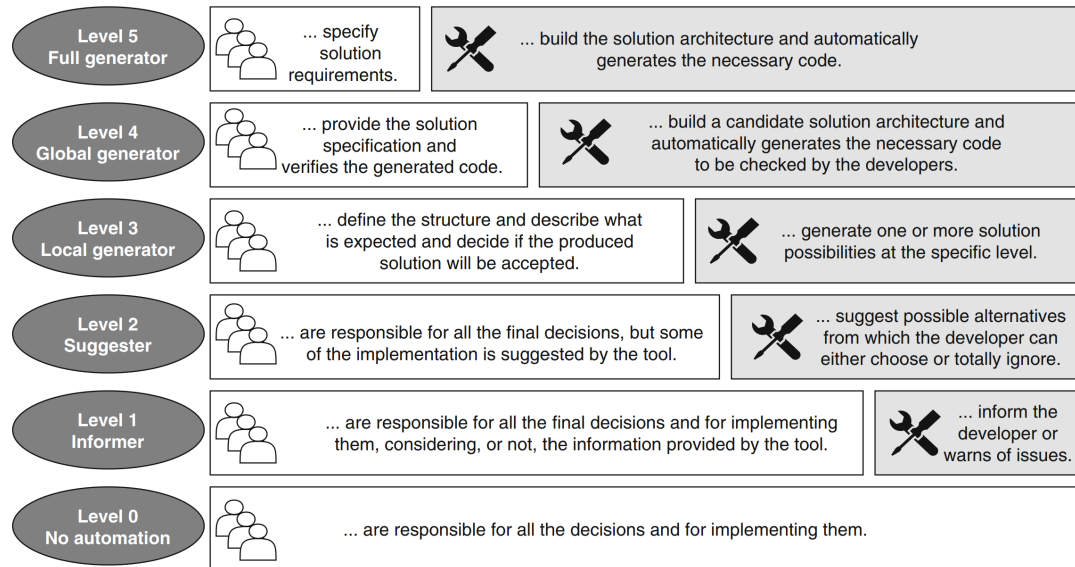


Figure 2.5: Degree of Automation Taxonomy for software Engineering
- six levels of automation according to DAnTE. (Melegati and Guerra 2024).

Trustworthy AI (High-Level Expert Group on Artificial Intelligence 2019), its principles:

- Human agency and oversight
- Robustness and safety
- Privacy and data governance
- Transparency
- Diversity, non-discrimination and fairness
- Societal and environmental well-being
- Accountability

Lastly, heavily recommended by the professor, the AI4People “An Ethical Framework for a Good AI Society” (Floridi 2019a) was another useful source, as it is also familiar since it bases its principles on bioethics, which were studied in the previous assignment, with the addition of a new principle, explicability.

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Altogether: (1) Beneficence, (2) Non-maleficence, (3) Autonomy, (4) Justice and (5) Explicability.

Because the principles seem to overlap, I decided to display the mapping of my own principles to these using a table 2.1. I steered away from what I would honestly consider somewhat obscured titles to be very explicit about what exactly each principle is about, but the relationships, I believe, are clear.

2.4 Practising engineering ethics

Regulation pushes industry to find frameworks, certifications and tools to apply and comply with it.

2.4.1 EU AI Act

The EU AI Act attempts to regulate AI systems. Here is how the system is defined in the Act:

“a machine-based system that is designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment, and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments” (European Parliament and Council of the European Union 2024a, Art. 3(1), p. 46).

Golpayegani et al. (2026), created a semantic framework based on the Act with great visuals drawn from it; Fig. 2.6 is incredibly useful to quickly transmit concepts surrounding the AI systems.

Intelligent Transport Systems that use AI for automated decision-making are in the domain of *critical infrastructure*, which are considered high-risk Fig. 2.7. And they will require documentation such as:

- Technical Documentation (European Parliament and Council of the European Union 2024a, Art. 11, Annex IV)

Code pillar	IBM AI Design Ethics (International Business Machines Corporation 2022)	Ethics Guidelines for Trustworthy AI (High-Level Expert Group on Artificial Intelligence 2019)	AI4People (Floridi 2019a)
Humans go first	Value alignment	Human agency and oversight	Autonomy
Human-thought	Explainability	Transparency; human agency and oversight	Explicability; autonomy
Help anyone you can in any way you can	Fairness; value alignment	Societal and environmental well-being	Beneficence
Comply with every rule and regulation	Accountability; user data rights	Privacy and data governance; accountability; robustness and safety	Justice; non-maleficence
Documentation	Accountability; explainability	Transparency; accountability	Explicability
Version control	Accountability	Robustness and safety; transparency; accountability	Explicability; non-maleficence
Problem-solving	Value alignment; explainability	Human agency and oversight; robustness and safety	Beneficence; non-maleficence
Scalability	Accountability; fairness	Robustness and safety; societal and environmental well-being	Justice; non-maleficence

Table 2.1: Mapping of the proposed code of conduct pillars to the reviewed AI ethics frameworks.

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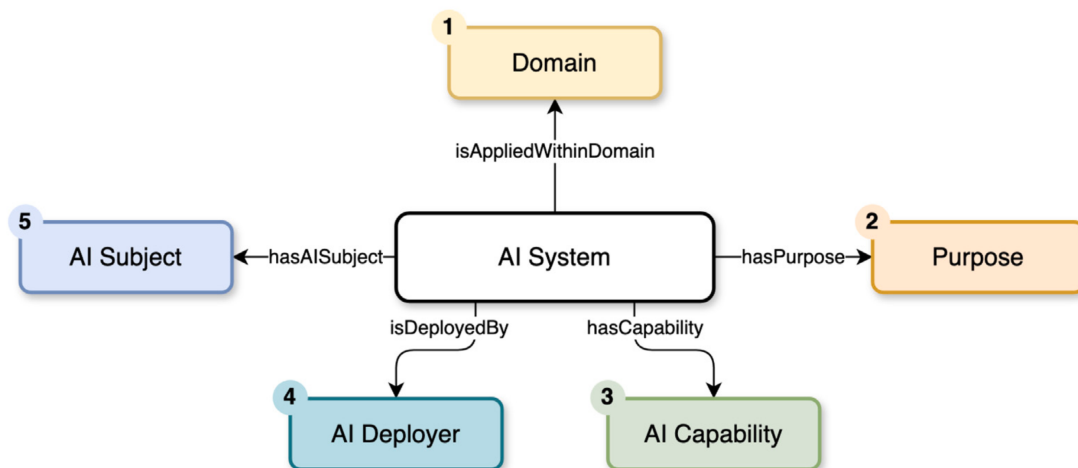


Figure 2.6: EU AI Act Annex III - concepts for determining high-risk AI systems (Golpayegani et al. 2026).

- Risk management system documentation (European Parliament and Council of the European Union 2024a, Art. 9)
- Instruction for use (European Parliament and Council of the European Union 2024a, Art.13)
- Post-market monitoring system documentation (European Parliament and Council of the European Union 2024a, Art.13)
- Post-market monitoring plan (European Parliament and Council of the European Union 2024a, Art.13)
- EU declaration of conformity (European Parliament and Council of the European Union 2024a, Art.18)

And being registered in the EU database for high-risk AI systems. This is why ‘Documentation’ is the first principle I mentioned in section 1.3. But more importantly, why I strongly suggest not to steer away from human decision-making. However, engineers who familiarise themselves with all the principles will know as well that even suggesting human options to respond to an incident (which would be genuinely useful if well-implemented) would also fall in the same category:

“AI system referred to in Annex III shall not be considered to be high-risk where it does not pose a significant risk of harm to the health, safety or fundamental rights of natural persons, including by not materially **influencing the outcome of decision making**” (European Parliament and Council of the European Union 2024a).

It is also important to notice that technological solutions often evolve and add features after deployment; for example, ITS infrastructure might allow the progression towards Cooperative Intelligent Transport Systems (C-ITS), which could create more space for risks that computer engineers will need to account for. This is why I asked readers to consider the code as a living document and the reason why abiding by the code is not enough to ethically comply with the Act; the responsibility ultimately lies in every developer.

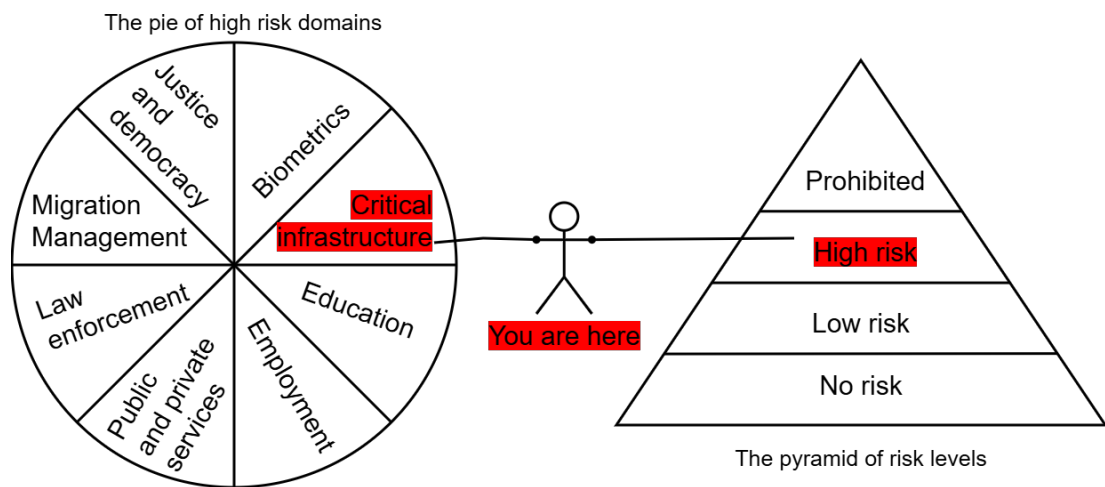


Figure 2.7: AI and ITS - you can easily find yourself in this place, in terms of the Act, if you mix both of them.

2.4.2 Beyond the EU AI Act

Intelligent traffic systems and AI-assisted engineering are not governed by the EU AI Act alone. Other legal, technical, and governance instruments also shape responsible engineering practice.

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GDPR The General Data Protection Regulation seeks to protect the privacy of individuals who might interact with ITS. It is just as important as the EU AI Act in the *critical infrastructure* domain because road users must be aware of when CCTV is in operation, and it demands that any personal data processed via an AI tool abides by the GDPR Article 6 (European Parliament and Council of the European Union 2016).

DATEX II Variable Message Signs (VMS) are a well-known piece of equipment that any ITS will make use of. They are extremely regulated; signs should only display lawful pictograms. “DATEX II is the reference data standard in Europe for road traffic and travel information” (DATEX II 2026).

CRA Devices that form the infrastructure of ITS will be scattered around motorways all around the country. Malicious acts for a variety of reasons might try to harm the system throughout a diverse range of attacks. Now more than ever, it is important to correctly harden devices. The Cyber Resilience Act “lays down essential cybersecurity requirements for the design, development and production of products with digital elements” (European Parliament and Council of the European Union 2024b).

None of these can be neglected by designers of ITS. Although not explicitly, my code of conduct tries to create a framework to keep engineers within the bounds of what is right.

2.4.3 Whistle-blowing

The European study: ‘A Governance Framework for Algorithmic Accountability and Transparency’ (European Parliamentary Research Service 2019) highlights the importance of whistleblowers as a pillar for raising awareness. I find awareness perhaps the most important factor in today’s landscape to reach a good AI society, because it is the first step to achieving the accountability that many codes of ethics are trying to pursue.

A considerable amount of time was spent reviewing and analysing different whistle-blowing protection mechanisms in organisations; this seems to be a must-have in most IT companies’ codes of conduct; HP calls it “How to raise a concern”

(HP Inc. 2025), Nokia “Speaking Up” (Nokia 2025) and IBM “Speaking Up – Where and How to Report” (International Business Machines Corporation n.d.). HP’s in particular uses an external tool for reporting (Convercent), this made me research other platforms, such as Navex, however, both need the company to hire the services.

Luckily, having a secure and anonymous channel to raise ethical concerns regarding AI is a national requirement (Workplace Relations Commission 2026), so I picked this as the last resort a computer engineer might use in case nothing else works, but I prioritised speaking with the closest manager or other management figures before because I believe collaboration should be pursued when possible.

2.5 Challenges

I must admit that despite trying my best for every assignment on our Ethics module, I still feel like a fish out of water trying to concoct a concise and valuable code of conduct. I often found myself trying to teach or educate my audience on matters related to EU AI Act compliance, GDPR compliance, or technical matters such as which specific tools or environments a computer engineer should use.

I was not underestimating my peers, but perhaps subconsciously through my readings and our lectures I realised that many professionals in the field are not trying to be unethical but are simply not even aware of the potential infractions committed on a daily basis. My eagerness to inform constantly got in the way of designing and communicating the principles computer engineers effectively should use to make their behaviours consistent with an ethical framework, instead of trying to hold their hands and explain details of implementation or regulation. Only after analysing a variety of codes of conduct for different professions and also IT companies did I manage to understand better the task at hand, but it was challenging, and some of my vices, despite my efforts, could still show.

From all the codes studied and reviewed, not a single one has an extension remotely close to the word count limit or page count limit assigned to the task. Valid pedagogic and logistic reasons are behind this, of that I am sure; not because it is a challenge means it is something bad by default. But even within a reduced

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space (focusing in critical infrastructure in the context of ITS), some aspects were sadly left unexplored, such as more on self-reporting (if you are the one who committed an unethical action) or genuinely useful technical frameworks like the LLM Wiki by Andrej Karpathy (Karpathy 2026).

The internal AI guidelines in my company (still work in progress) currently consist of 21 pages. And from my research space, (Association for Computing Machinery 2018) is the smallest code of them all, still more than double the content (word count) that the assignment suggested. It is important to clarify that I understand these were soft limits, but every extra piece of content in the code of conduct would also need to expand the analysis for part B, deterring me from moving further away from the bounds.

Finally, the biggest challenge was not to fall into any of the pitfalls highlighted by (Floridi 2019b). Ethics-shopping and ethics-washing were constantly in my mind as practices I needed to avoid at all costs. It is a strange but real dichotomy on which one can fall if, while creating a code of conduct for ethical usage of a tool, one becomes unethical.

2.6 Final thoughts

State-of-the-art theory about codes of conduct focuses on creating the right environment so that individuals avoid the *moral dissonance* that the difference between *what is right* and *what is wrong* affects the minimum individuals in critical positions. Although the EU AI Act is a step in the right direction, I believe it does not acknowledge a problem that could eventually get the best of humanity: skill atrophy. I would never claim that my code of conduct is flawless, but it tries to address the importance of professional and human excellence in a high-risk domain such as Intelligent Traffic Systems.

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